

Topology of the real line – 2

Limit points, perfect sets and Cantor sets

Homework

8 сентября 2026 года

Theoretical minimum. Limit points, isolated points and condensation points; the derived set and its closedness; perfect sets and the Cantor–Bendixson theorem; a non-empty perfect set has cardinality continuum; the middle-thirds set, its ternary description and its self-similarity; Cantor-type sets of positive measure.

1. (Karasev 1.2) Let A be an uncountable set of real numbers. Prove that there is a strictly increasing sequence all of whose terms belong to A . Prove also that A contains one of its own condensation points, and that all but countably many points of A are of this kind.
2. Construct a closed nowhere dense subset $K \subseteq [0, 1]$ of Lebesgue measure at least $\frac{1}{2}$, and prove both properties. Its complement in $[0, 1]$ is then open and dense of measure at most $\frac{1}{2}$: describe its components explicitly and compute the sum of their lengths.
3. (IMC) Let $f: [a, b] \rightarrow [a, b]$ be continuous, let $p \in [a, b]$ and let $p_0 = p$, $p_{n+1} = f(p_n)$. Suppose the set $T_p = \{p_n : n \geq 0\}$ is closed. Show that T_p is finite.
4. Prove that a set $F \subseteq \mathbb{R}$ is the derived set of some set if and only if F is closed, and that F is the set of limit points of some sequence if and only if F is closed. Construct a sequence whose set of limit points is exactly the Cantor set.
5. Prove that C is homeomorphic to $\{0, 1\}^{\mathbb{N}}$ and to $C \times C$, and deduce that C is homeomorphic to a nowhere dense proper subset of itself. Explain why no interval has this property.

6. (Putnam 1990) Consider a paper punch which can be centred at any point of the plane and, when operated, removes precisely those points whose distance from the centre is irrational. How many punches are needed to remove every point of the plane?
7. (IMC) Let $f: (0, 1) \rightarrow [0, \infty)$ vanish except at the distinct points $a_1, a_2, \dots$, and put $b_n = f(a_n)$. Prove that if $\sum_n b_n < \infty$, then f is differentiable at at least one point of $(0, 1)$.
8. Prove that the endpoints of the removed intervals form a countable dense subset of C , and that C minus this set is still uncountable. Which points of C are ternary rationals?
9. (MIPT) In how many ways — in the sense of cardinality — can the number 1 be written as a finite or infinite sum of a subset of $\{\varphi^{-n} : n \in \mathbb{Z}_+\}$, where $\varphi = \frac{1+\sqrt{5}}{2}$?
10. Prove that a countable closed subset of $\mathbb{R}$ has an isolated point, and deduce that $\mathbb{Q}$ is not closed in a strong sense: no countable closed set can be dense in an interval. Then give an example of a countable closed set whose derived set is infinite.